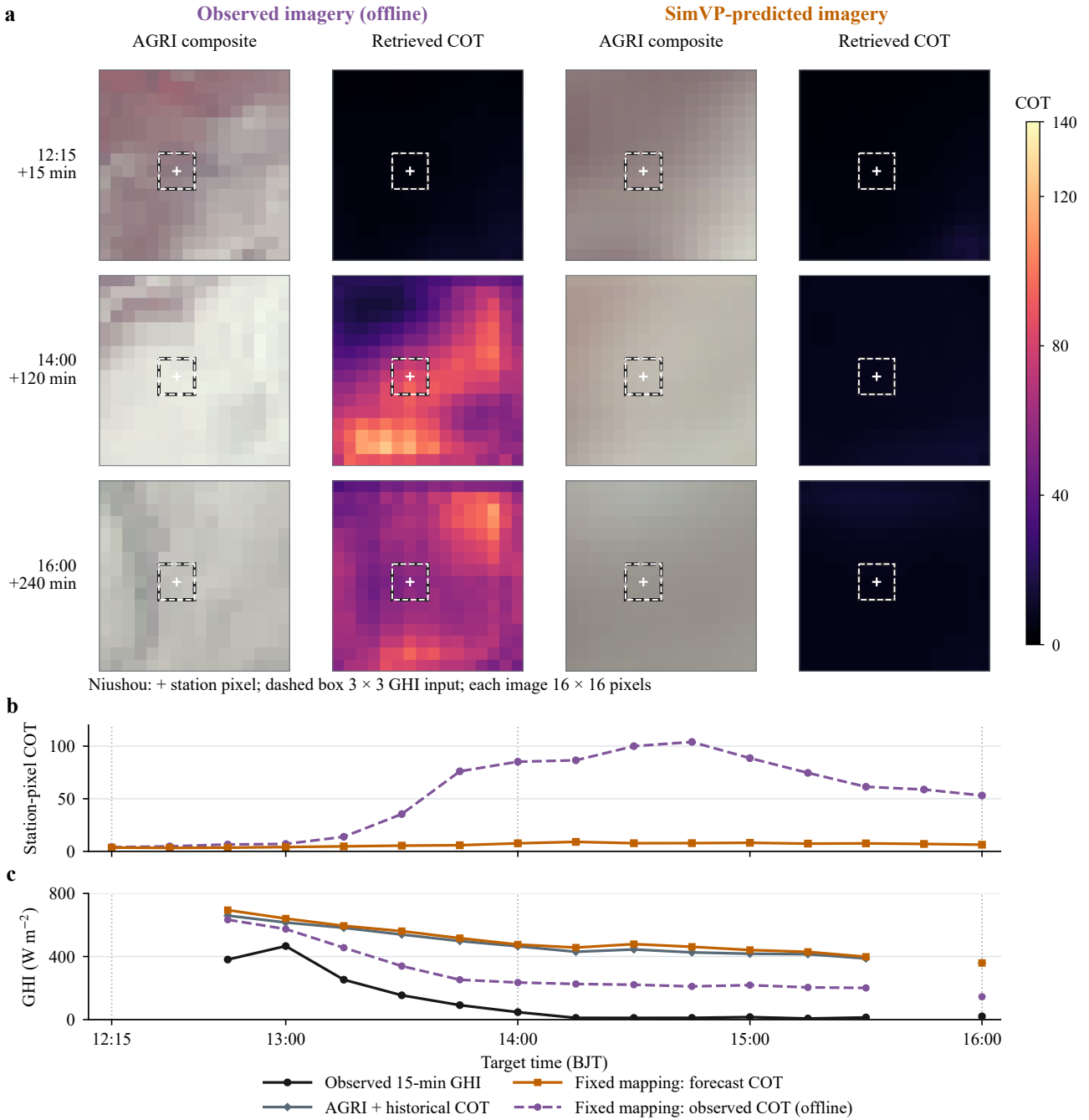

a
Observed imagery (offline)
SimVP-predicted imagery
AGRI composite
Retrieved COT
AGRI composite
Retrieved COT
12:15
+15 min
14:00
+120 min
16:00
+240 min
COT
140
120
80
40
0
Niushou: + station pixel; dashed box 3 × 3 GHI input; each image 16 × 16 pixels
b
Station-pixel COT
100
50
0
c
GHI (W m−2)
800
400
0
12:15
13:00
14:00
15:00
16:00
Target time (BJT)
Observed 15-min GHI
Fixed mapping: forecast COT
AGRI + historical COT
Fixed mapping: observed COT (offline)
6 August 2025 | initialization 12:00 BJT | calendar-selected development OOF case